

VI LENA Supplement

Erin Campbell

2025-01-21

Equivalence Testing

```
# Adult Word Count: d=0.3
```

```
TOSTER::t_TOST(AWC_per_hour ~ group, data = LENA_counts, hypothesis = "EQU", paired = TRUE,  
               var.equal = TRUE, eqb = .3, eqbound_type = "SMD", alpha = .05)
```

```
##  
## Paired t-test  
##  
## The equivalence test was non-significant, t(14) = 1.14, p = 0.14  
## The null hypothesis test was non-significant, t(14) = -0.0202p = 0.98  
## NHST: don't reject null significance hypothesis that the effect is equal to zero  
## TOST: don't reject null equivalence hypothesis  
##  
## TOST Results  
##  
##           t df p.value  
## t-test    -0.02022 14  0.984  
## TOST Lower  1.14168 14  0.136  
## TOST Upper -1.18211 14  0.128  
##  
## Effect Sizes  
##           Estimate      SE      C.I. Conf. Level  
## Raw          -3.630515 179.5779 [-319.9229, 312.6619]      0.9  
## Hedges's g(z) -0.004934  0.2582  [-0.4063, 0.3966]      0.9  
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Adult Word Count: d=0.5
```

```
TOSTER::t_TOST(AWC_per_hour ~ group, data = LENA_counts, hypothesis = "EQU", paired = TRUE,  
               var.equal = TRUE, eqb = .5, eqbound_type = "SMD", alpha = .05)
```

```
##  
## Paired t-test  
##  
## The equivalence test was significant, t(14) = 1.92, p = 0.04  
## The null hypothesis test was non-significant, t(14) = -0.0202p = 0.98  
## NHST: don't reject null significance hypothesis that the effect is equal to zero  
## TOST: reject null equivalence hypothesis  
##  
## TOST Results
```

```
##               t df p.value
## t-test      -0.02022 14  0.984
## TOST Lower  1.91627 14  0.038
## TOST Upper -1.95671 14  0.035
##
## Effect Sizes
##               Estimate      SE      C.I. Conf. Level
## Raw          -3.630515 179.5779 [-319.9229, 312.6619]      0.9
## Hedges's g(z) -0.004934  0.2582  [-0.4063, 0.3966]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Adult Word Count: d=0.7
```

```
TOSTER::t_TOST(AWC_per_hour ~ group, data = LENA_counts, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was significant, t(14) = 2.69, p < 0.01
## The null hypothesis test was non-significant, t(14) = -0.0202p = 0.98
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##               t df p.value
## t-test      -0.02022 14  0.984
## TOST Lower  2.69087 14  0.009
## TOST Upper -2.73131 14  0.008
##
## Effect Sizes
##               Estimate      SE      C.I. Conf. Level
## Raw          -3.630515 179.5779 [-319.9229, 312.6619]      0.9
## Hedges's g(z) -0.004934  0.2582  [-0.4063, 0.3966]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Manual Word Count: d=0.3
```

```
TOSTER::t_TOST(MWC_per_hour ~ group, data = manual_word_tokens, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .3, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = -0.1, p = 0.46
## The null hypothesis test was non-significant, t(14) = 1.06p = 0.31
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##               t df p.value
## t-test       1.0600 14  0.307
## TOST Lower  2.2219 14  0.022
## TOST Upper -0.1019 14  0.460
##
```

```
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw      301.7333 284.6477 [-199.6195, 803.0862]      0.9
## Hedges's g(z) 0.2587 0.2634 [-0.155, 0.6635]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").

# Manual Word Count: d=0.5
TOSTER::t_TOST(MWC_per_hour ~ group, data = manual_word_tokens, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .5, eqbound_type = "SMD", alpha = .05)

##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = -0.88, p = 0.2
## The null hypothesis test was non-significant, t(14) = 1.06p = 0.31
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      1.0600 14 0.307
## TOST Lower  2.9965 14 0.005
## TOST Upper -0.8765 14 0.198
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw      301.7333 284.6477 [-199.6195, 803.0862]      0.9
## Hedges's g(z) 0.2587 0.2634 [-0.155, 0.6635]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").

# Manual Word Count: d=0.7
TOSTER::t_TOST(MWC_per_hour ~ group, data = manual_word_tokens, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)

##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = -1.7, p = 0.06
## The null hypothesis test was non-significant, t(14) = 1.06p = 0.31
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      1.060 14 0.307
## TOST Lower  3.771 14 0.001
## TOST Upper -1.651 14 0.060
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw      301.7333 284.6477 [-199.6195, 803.0862]      0.9
## Hedges's g(z) 0.2587 0.2634 [-0.155, 0.6635]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Conversational Turn Count: d=0.3
TOSTER::t_TOST(CTC_per_hour ~ group, data = LENA_counts, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .3, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = -0.74, p = 0.23
## The null hypothesis test was non-significant, t(14) = 0.418p = 0.68
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.4176 14  0.683
## TOST Lower  1.5795 14  0.068
## TOST Upper -0.7443 14  0.235
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           2.5570 6.123 [-8.2267, 13.3407]      0.9
## Hedges's g(z)  0.1019 0.259 [-0.3025, 0.5028]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Conversational Turn Count: d=0.5
TOSTER::t_TOST(CTC_per_hour ~ group, data = LENA_counts, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .5, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = -1.5, p = 0.08
## The null hypothesis test was non-significant, t(14) = 0.418p = 0.68
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.4176 14  0.683
## TOST Lower  2.3541 14  0.017
## TOST Upper -1.5188 14  0.076
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           2.5570 6.123 [-8.2267, 13.3407]      0.9
## Hedges's g(z)  0.1019 0.259 [-0.3025, 0.5028]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Conversational Turn Count: d=0.7
TOSTER::t_TOST(CTC_per_hour ~ group, data = LENA_counts, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was significant,  $t(14) = -2.3$ ,  $p = 0.02$ 
## The null hypothesis test was non-significant,  $t(14) = 0.418$ ,  $p = 0.68$ 
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.4176 14  0.683
## TOST Lower  3.1287 14  0.004
## TOST Upper -2.2934 14  0.019
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           2.5570 6.123 [-8.2267, 13.3407]      0.9
## Hedges's g(z)  0.1019 0.259 [-0.3025, 0.5028]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Proportion of Child-Directed Speech: d=0.3
TOSTER::t_TOST(prop_CDS ~ group, data = xds_props_wide, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .3, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant,  $t(14) = -0.92$ ,  $p = 0.19$ 
## The null hypothesis test was non-significant,  $t(14) = 0.244$ ,  $p = 0.81$ 
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.2439 14  0.811
## TOST Lower  1.4058 14  0.091
## TOST Upper -0.9180 14  0.187
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           0.01632 0.06693 [-0.1016, 0.1342]      0.9
## Hedges's g(z)  0.05953 0.25848 [-0.3434, 0.4604]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Proportion of Child-Directed Speech: d=0.5
TOSTER::t_TOST(prop_CDS ~ group, data = xds_props_wide, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .5, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant,  $t(14) = -1.7$ ,  $p = 0.06$ 
## The null hypothesis test was non-significant,  $t(14) = 0.244$ ,  $p = 0.81$ 
```

```
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.2439 14  0.811
## TOST Lower  2.1804 14  0.023
## TOST Upper -1.6926 14  0.056
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           0.01632 0.06693 [-0.1016, 0.1342]      0.9
## Hedges's g(z) 0.05953 0.25848 [-0.3434, 0.4604]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Proportion of Child-Directed Speech: d=0.7
TOSTER::t_TOST(prop_CDS ~ group, data = xds_props_wide, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was significant, t(14) = -2.5, p = 0.01
## The null hypothesis test was non-significant, t(14) = 0.244p = 0.81
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.2439 14  0.811
## TOST Lower  2.9550 14  0.005
## TOST Upper -2.4672 14  0.014
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           0.01632 0.06693 [-0.1016, 0.1342]      0.9
## Hedges's g(z) 0.05953 0.25848 [-0.3434, 0.4604]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Type-Token Ratio: d=0.3
TOSTER::t_TOST(mean_ttr ~ group, data = TTR_calculations, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .3, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = 0.2, p = 0.42
## The null hypothesis test was non-significant, t(14) = -0.96p = 0.35
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
```

```
## t-test      -0.9597 14    0.353
## TOST Lower  0.2022 14    0.421
## TOST Upper -2.1216 14    0.026
##
## Effect Sizes
##           Estimate      SE          C.I. Conf. Level
## Raw          -0.01233 0.01285  [-0.035, 0.0103]      0.9
## Hedges's g(z) -0.23423 0.26249  [-0.6381, 0.1777]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Type-Token Ratio: d=0.5
TOSTER::t_TOST(mean_ttr ~ group, data = TTR_calculations, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .5, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = 0.98, p = 0.17
## The null hypothesis test was non-significant, t(14) = -0.96p = 0.35
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      -0.9597 14    0.353
## TOST Lower   0.9768 14    0.173
## TOST Upper  -2.8962 14    0.006
##
## Effect Sizes
##           Estimate      SE          C.I. Conf. Level
## Raw          -0.01233 0.01285  [-0.035, 0.0103]      0.9
## Hedges's g(z) -0.23423 0.26249  [-0.6381, 0.1777]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Type-Token Ratio: d=0.7
TOSTER::t_TOST(mean_ttr ~ group, data = TTR_calculations, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was non-significant, t(14) = 1.75, p = 0.05
## The null hypothesis test was non-significant, t(14) = -0.96p = 0.35
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: don't reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      -0.9597 14    0.353
## TOST Lower   1.7514 14    0.051
## TOST Upper  -3.6708 14    0.001
##
## Effect Sizes
```

```
##           Estimate      SE           C.I. Conf. Level
## Raw      -0.01233 0.01285 [-0.035, 0.0103]          0.9
## Hedges's g(z) -0.23423 0.26249 [-0.6381, 0.1777]          0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Proportion of Highly Visual Words: d=0.3
```

```
TOSTER::t_TOST(prop_Visual ~ group, data = sensory_props_wide, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was significant, t(14) = -1.9, p = 0.04
## The null hypothesis test was non-significant, t(14) = 0.778p = 0.45
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.7776 14  0.450
## TOST Lower  3.4887 14  0.002
## TOST Upper -1.9335 14  0.037
##
## Effect Sizes
##           Estimate      SE           C.I. Conf. Level
## Raw          0.01399 0.01799 [-0.0177, 0.0457]          0.9
## Hedges's g(z) 0.18980 0.26103 [-0.2192, 0.5922]          0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

```
# Proportion of Highly Visual Words: d=0.5
```

```
TOSTER::t_TOST(prop_Visual ~ group, data = sensory_props_wide, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

```
##
## Paired t-test
##
## The equivalence test was significant, t(14) = -1.9, p = 0.04
## The null hypothesis test was non-significant, t(14) = 0.778p = 0.45
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.7776 14  0.450
## TOST Lower  3.4887 14  0.002
## TOST Upper -1.9335 14  0.037
##
## Effect Sizes
##           Estimate      SE           C.I. Conf. Level
## Raw          0.01399 0.01799 [-0.0177, 0.0457]          0.9
## Hedges's g(z) 0.18980 0.26103 [-0.2192, 0.5922]          0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

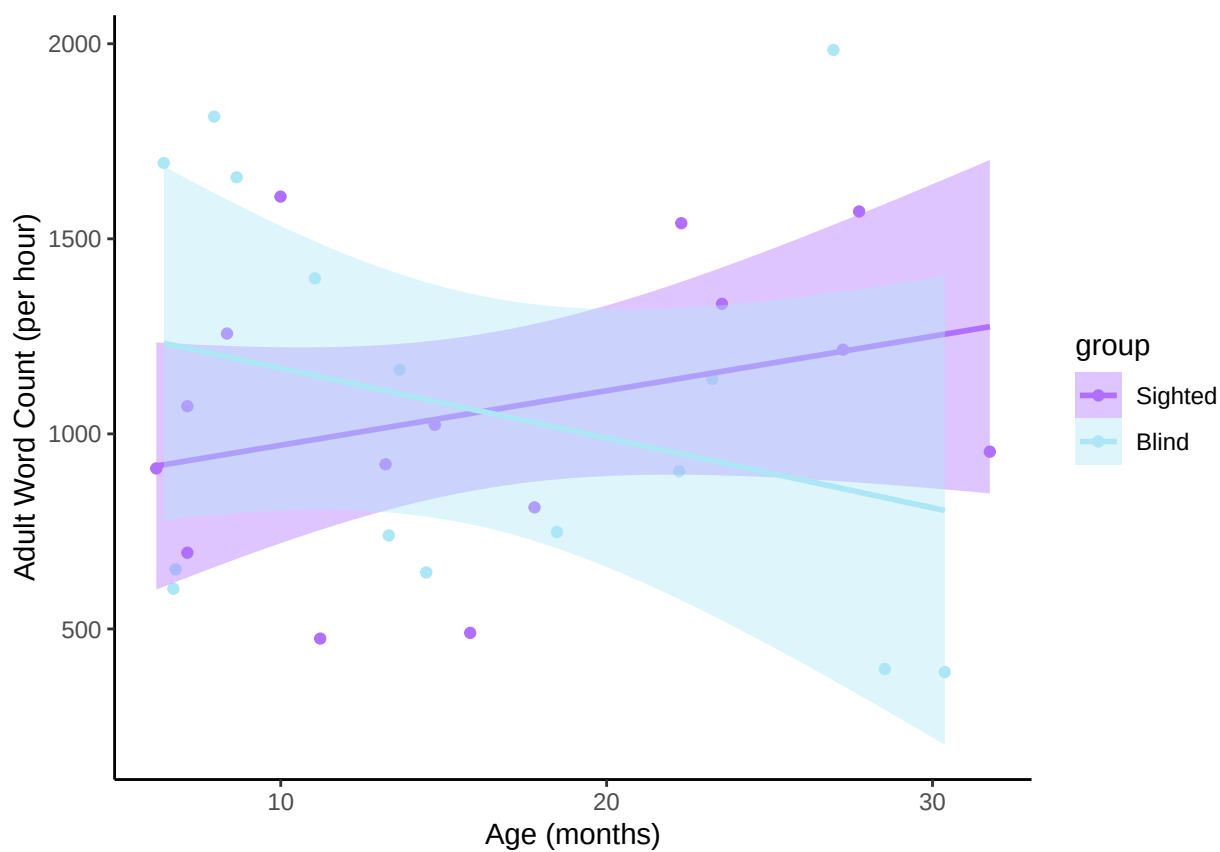


```
# Proportion of Highly Visual Words, d=0.7
TOSTER::t_TOST(prop_Visual ~ group, data = sensory_props_wide, hypothesis = "EQU", paired = TRUE,
               var.equal = TRUE, eqb = .7, eqbound_type = "SMD", alpha = .05)
```

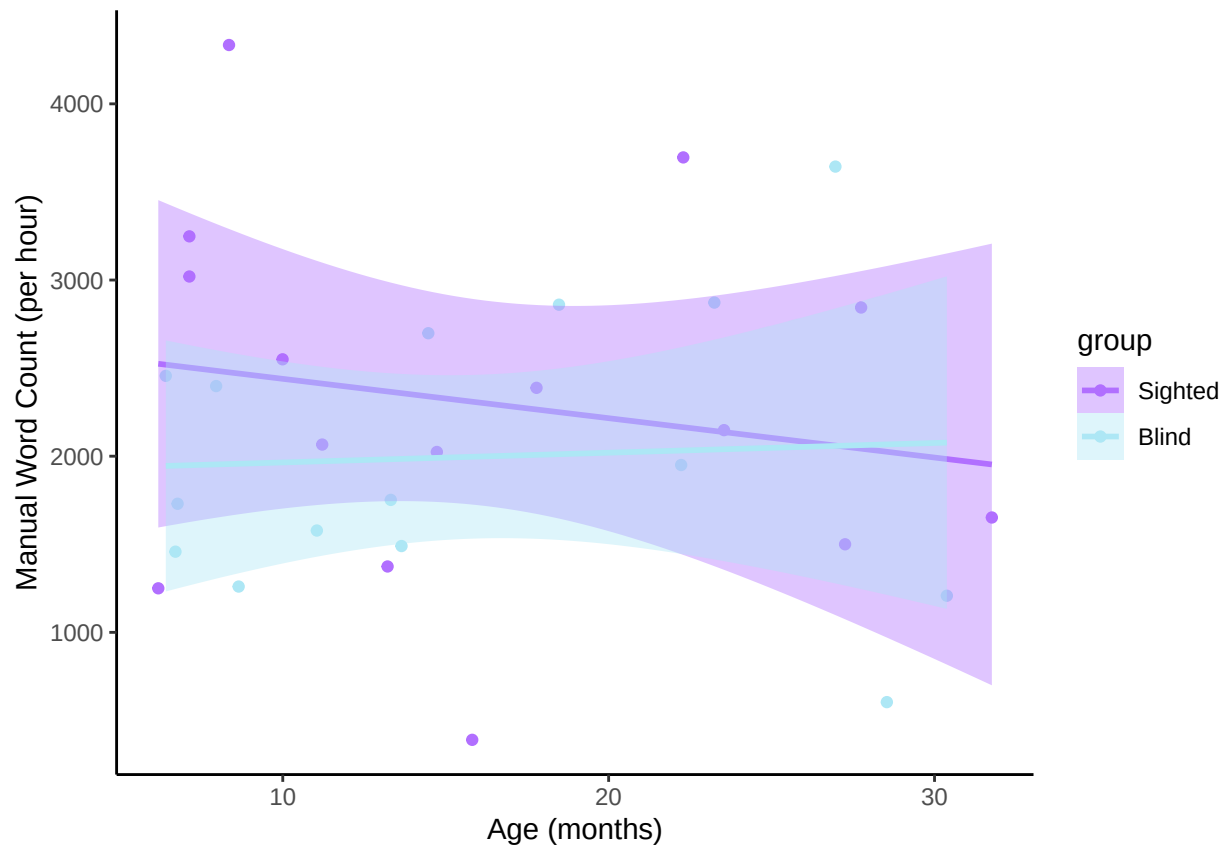
```
##
## Paired t-test
##
## The equivalence test was significant, t(14) = -1.9, p = 0.04
## The null hypothesis test was non-significant, t(14) = 0.778p = 0.45
## NHST: don't reject null significance hypothesis that the effect is equal to zero
## TOST: reject null equivalence hypothesis
##
## TOST Results
##           t df p.value
## t-test      0.7776 14  0.450
## TOST Lower  3.4887 14  0.002
## TOST Upper -1.9335 14  0.037
##
## Effect Sizes
##           Estimate      SE      C.I. Conf. Level
## Raw           0.01399 0.01799 [-0.0177, 0.0457]      0.9
## Hedges's g(z) 0.18980 0.26103 [-0.2192, 0.5922]      0.9
## Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

Age Interactions

Quantity

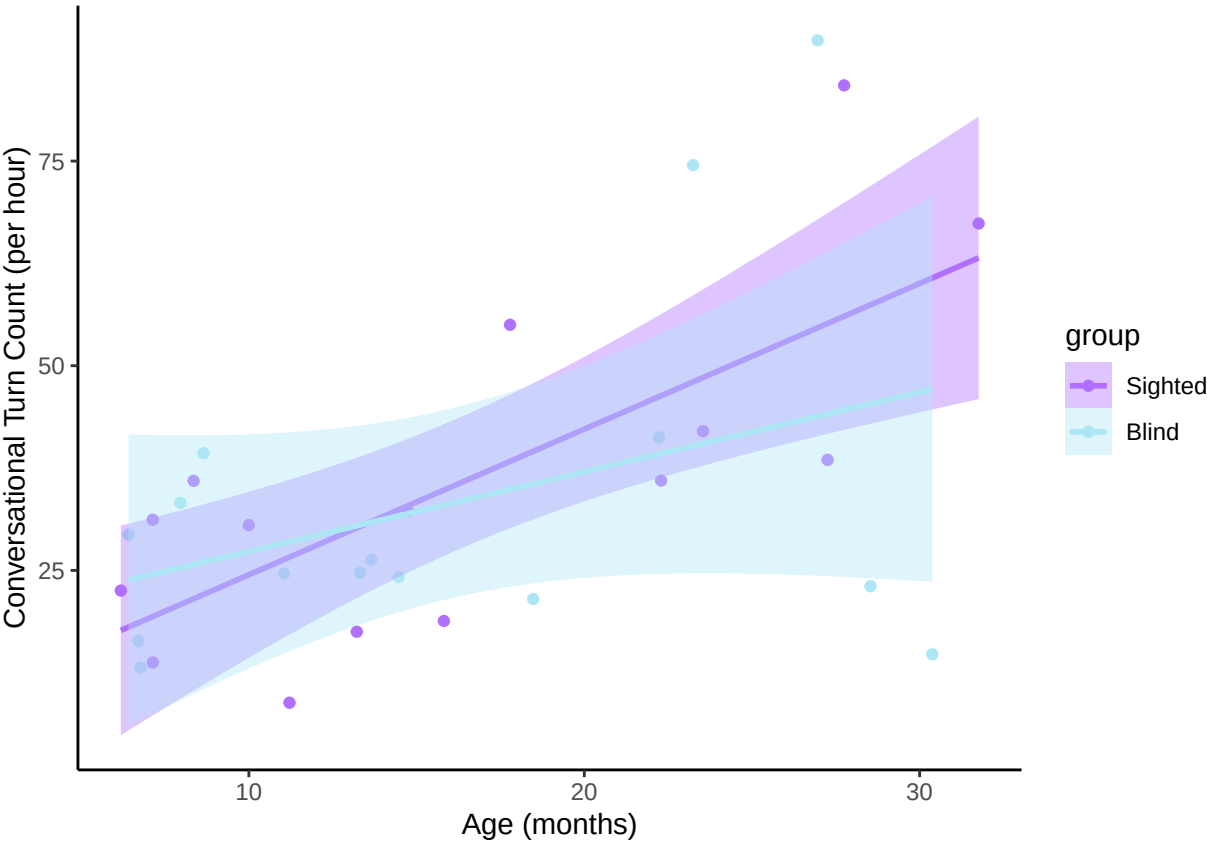


term	estimate	std.error	statistic	p.value
(Intercept)	831.32	260.89	3.19	.004
groupBlind	515.61	364.94	1.41	.170
Age_months	13.96	14.34	0.97	.339
groupBlind:Age_months	-31.85	20.22	-1.57	.127

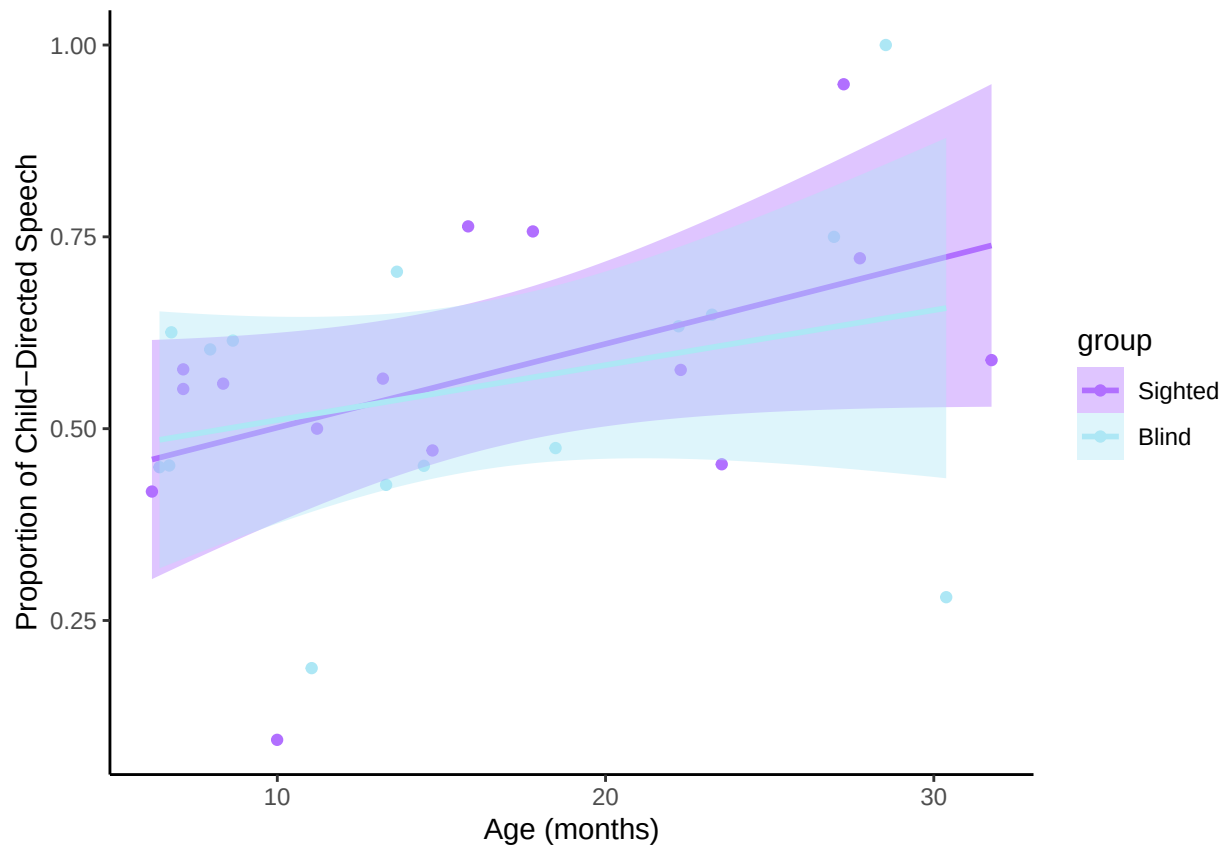


term	estimate	std.error	statistic	p.value
(Intercept)	2,662.53	545.85	4.88	< .001
groupBlind	-753.27	763.56	-0.99	.333
Age_months	-22.34	30.00	-0.74	.463
groupBlind:Age_months	27.86	42.31	0.66	.516

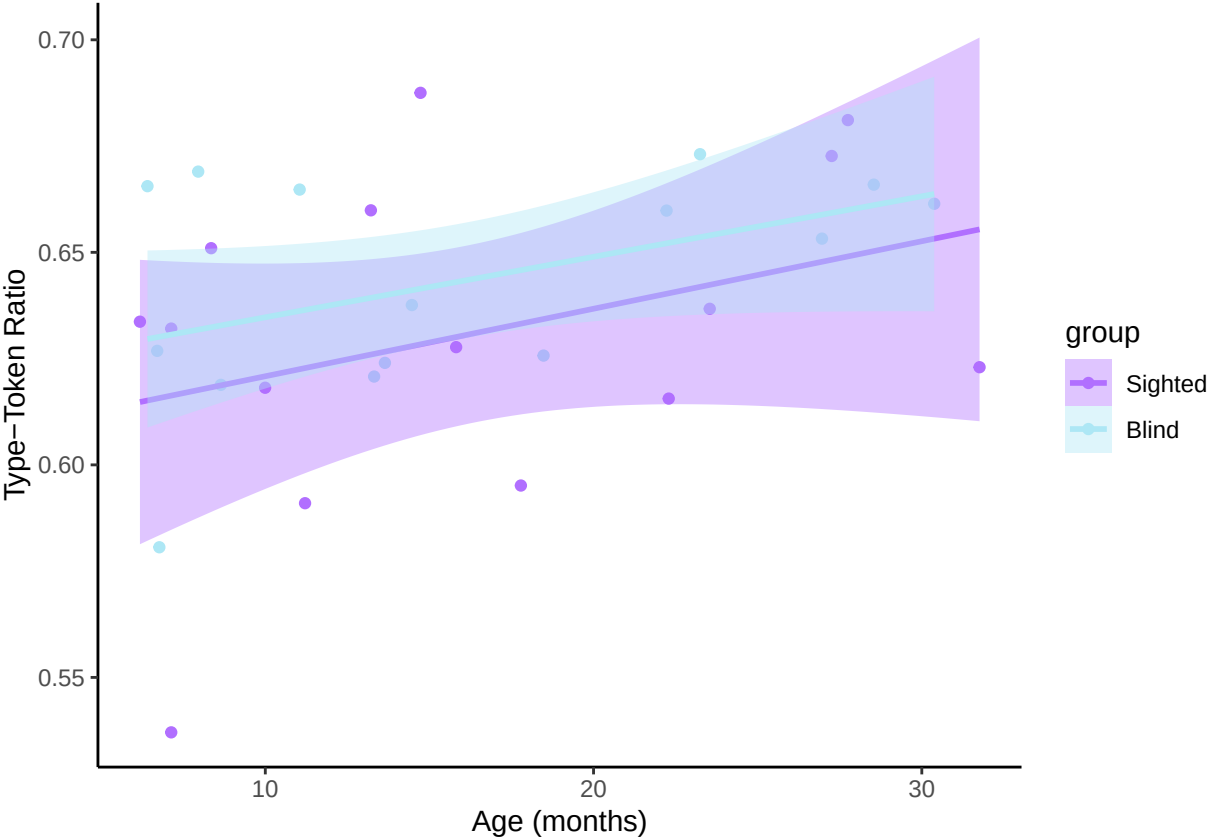
Interaction



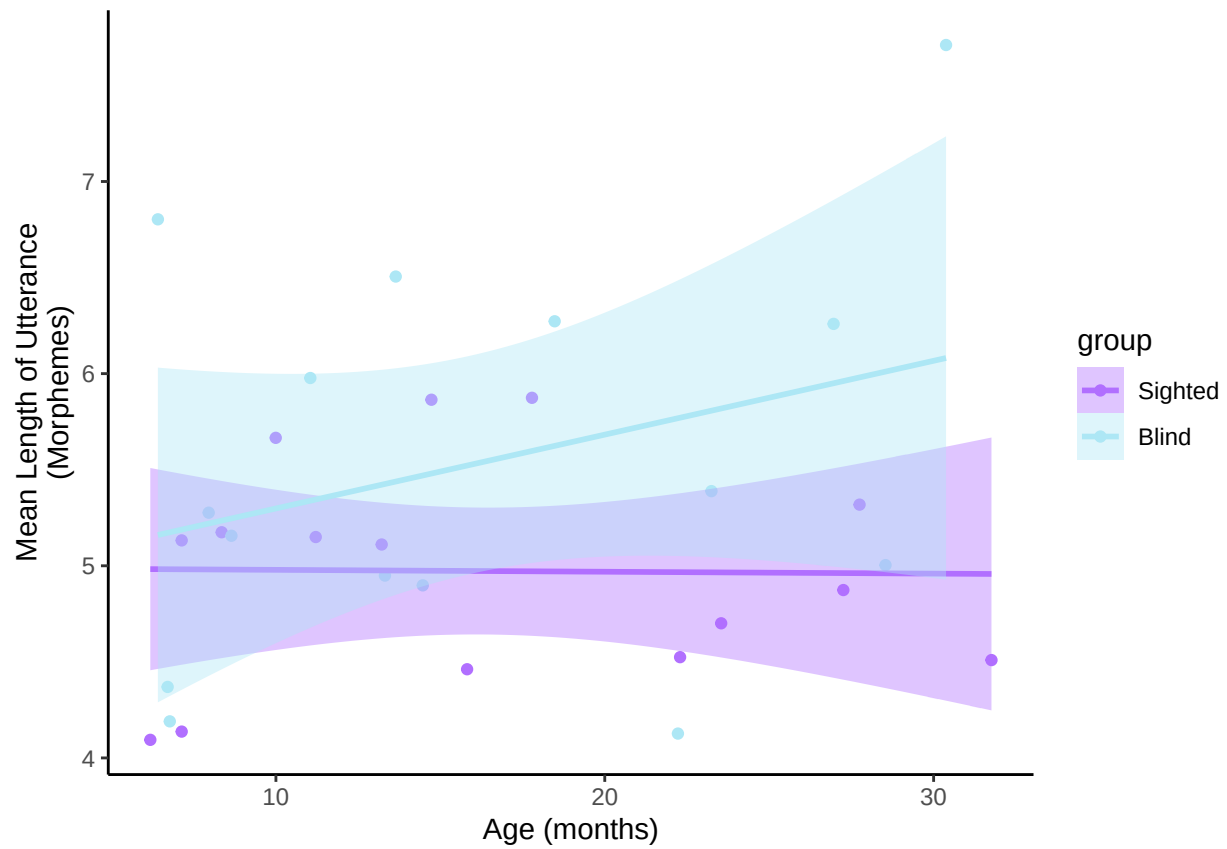
term	estimate	std.error	statistic	p.value
(Intercept)	6.67	10.32	0.65	.524
groupBlind	10.92	14.44	0.76	.456
Age__months	1.78	0.57	3.14	.004
groupBlind:Age__months	-0.81	0.80	-1.01	.323



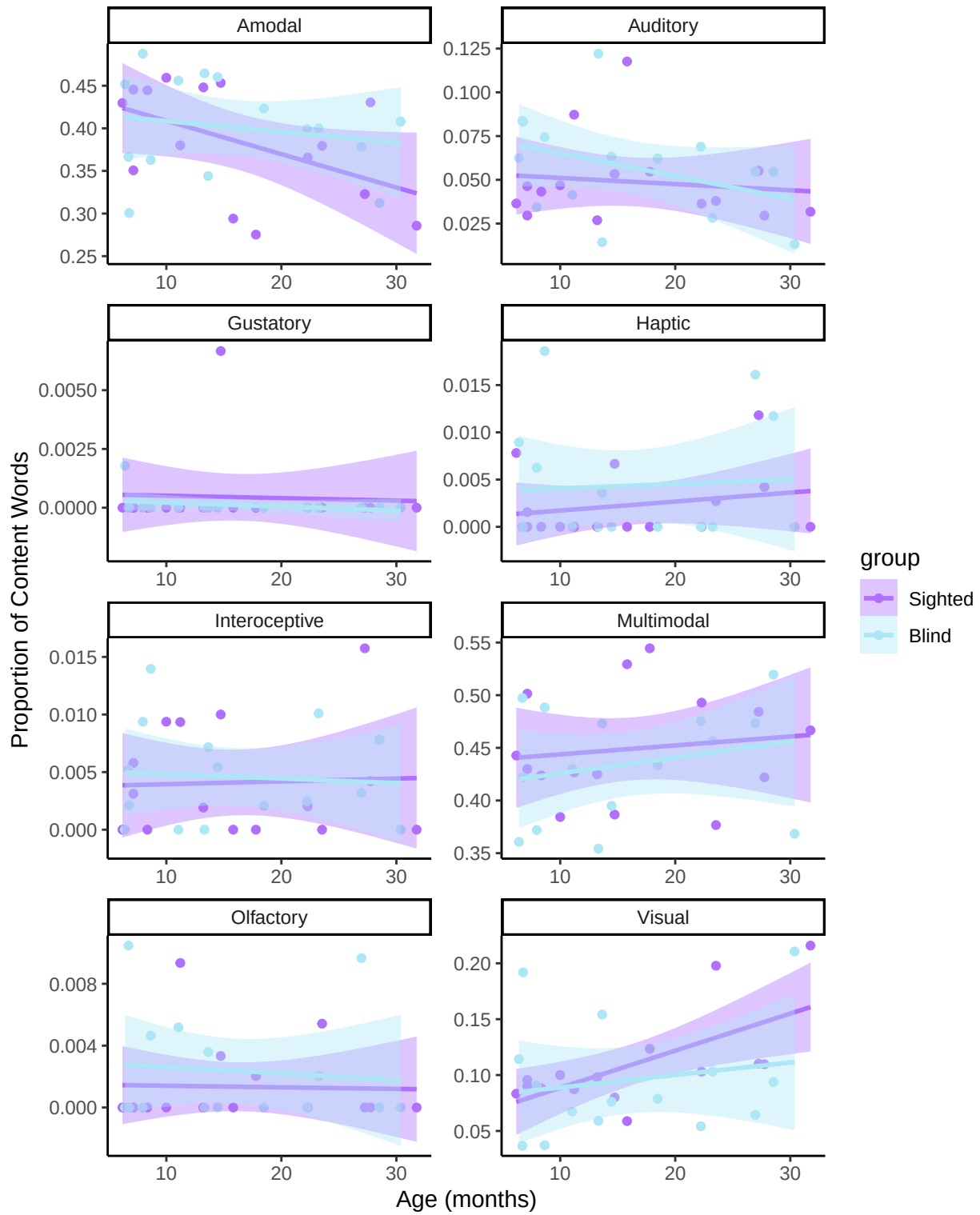
Linguistic



term	estimate	std.error	statistic	p.value
(Intercept)	0.60	0.02	33.05	< .001
groupBlind	0.02	0.03	0.61	.550
Age_months	0.00	0.00	1.58	.127
groupBlind:Age_months	0.00	0.00	-0.12	.908



Conceptual



term	estimate	std.error	statistic	p.value
(Intercept)	0.45	0.02	22.96	< .001
ModalityAuditory	-0.39	0.03	-14.26	< .001
ModalityGustatory	-0.45	0.03	-16.22	< .001
ModalityHaptic	-0.45	0.03	-16.21	< .001
ModalityInteroceptive	-0.44	0.03	-16.10	< .001
ModalityMultimodal	-0.01	0.03	-0.46	.645
ModalityOlfactory	-0.45	0.03	-16.18	< .001
ModalityVisual	-0.39	0.03	-14.23	< .001
groupBlind	-0.03	0.03	-0.98	.328
Age_months	0.00	0.00	-3.65	< .001
ModalityAuditory:groupBlind	0.05	0.04	1.30	.194
ModalityGustatory:groupBlind	0.03	0.04	0.69	.492
ModalityHaptic:groupBlind	0.03	0.04	0.77	.444
ModalityInteroceptive:groupBlind	0.03	0.04	0.73	.464
ModalityMultimodal:groupBlind	0.00	0.04	0.06	.954
ModalityOlfactory:groupBlind	0.03	0.04	0.73	.464
ModalityVisual:groupBlind	0.05	0.04	1.27	.206
ModalityAuditory:Age_months	0.00	0.00	2.35	.020
ModalityGustatory:Age_months	0.00	0.00	2.58	.011
ModalityHaptic:Age_months	0.00	0.00	2.64	.009
ModalityInteroceptive:Age_months	0.00	0.00	2.60	.010
ModalityMultimodal:Age_months	0.00	0.00	3.14	.002
ModalityOlfactory:Age_months	0.00	0.00	2.58	.011
ModalityVisual:Age_months	0.01	0.00	4.77	< .001
groupBlind:Age_months	0.00	0.00	1.75	.082
ModalityAuditory:groupBlind:Age_months	0.00	0.00	-1.68	.095
ModalityGustatory:groupBlind:Age_months	0.00	0.00	-1.24	.217
ModalityHaptic:groupBlind:Age_months	0.00	0.00	-1.26	.210
ModalityInteroceptive:groupBlind:Age_months	0.00	0.00	-1.27	.207
ModalityMultimodal:groupBlind:Age_months	0.00	0.00	-0.93	.353
ModalityOlfactory:groupBlind:Age_months	0.00	0.00	-1.25	.213
ModalityVisual:groupBlind:Age_months	0.00	0.00	-2.26	.025

